

Kindly upload & run in Sagemath

In [7]: `print ("Problem 9.1 part a")`

Out[7]: Problem 9.1 part a

In [3]:

```
# Heat Conduction from a Sphere - Part (a)
# Solve differential equation using desolve() - Symbolic Display
from sage.all import *

# Define symbolic variables
r = var('r', domain='positive')
a, T0, T_inf = var('a T_0 T_infty', domain='positive')
T = function('T')(r)
print("=" * 80)
print("HEAT CONDUCTION FROM A SPHERE")
print("=" * 80)

print("\nLaplace equation in spherical coordinates (radial symmetry):")
print("∇²T = 0")

print("\nIn radial form:")
print("(1/r²) d/dr(r² dT/dr) = 0")
print("\nMultiplying by r²:")
diff_eq = diff(r**2 * diff(T, r), r) == 0
print(f"d/dr(r² dT/dr) = 0")

print(f"\nDifferential equation to solve:")
show(diff_eq)
```

Out[3]: =====
HEAT CONDUCTION FROM A SPHERE
=====

Laplace equation in spherical coordinates (radial symmetry):
 $\nabla^2 T = 0$

In radial form:
 $(1/r^2) \frac{d}{dr}(r^2 \frac{dT}{dr}) = 0$

Multiplying by r^2 :
 $\frac{d}{dr}(r^2 \frac{dT}{dr}) = 0$

Differential equation to solve:

$$r^2 \frac{\partial^2}{(\partial r)^2} T(r) + 2r \frac{\partial}{\partial r} T(r) = 0$$

In [93]:

```
general_solution = desolve(diff_eq, T)

print(f"\n {T} = ")
```

```
show(general_solution)
```

Out[93]:

$$T(r) = K_1 + \frac{K_2}{r}$$

In [95]:

```
# Step 3: Apply boundary conditions
print("\n" + "Apply Boundary Conditions")
print("-" * 80)

print("\nBoundary Condition 1 (BC1):")
print("At r = a (sphere surface): T = T_o")
print(f"T(a) = T_o")

print("\nBoundary Condition 2 (BC2):")
print("As r → ∞ (far field): T → T_∞")
print(f"lim(r→∞) T(r) = T_∞")
```

Out[95]:

Apply Boundary Conditions

Boundary Condition 1 (BC1):
At r = a (sphere surface): T = T_o
T(a) = T_o

Boundary Condition 2 (BC2):
As r → ∞ (far field): T → T_∞
lim(r→∞) T(r) = T_∞

In [103]:

```
# Extract constants from general solution
C1, C2 = symbols('_K1 _K2', real=True)
print("\nGeneral form:")
print(f"T(r) = {C1} + {C2}/r")

print("\n--- From BC2 (r → ∞) ---")
print(f"lim(r→∞) [{C1} + {C2}/r] = T_∞")
print(f"{C2}/r → 0 as r → ∞")
print(f"Therefore: {C1} = T_∞")
C1_sol = T_inf
```

Out[103]:

General form:
T(r) = _K1 + _K2/r

--- From BC2 (r → ∞) ---
lim(r→∞) [_K1 + _K2/r] = T_∞
_K2/r → 0 as r → ∞
Therefore: _K1 = T_∞

In [111]:

```
print("\n--- From BC1 (r = a) ---")
print(f"At r = a: {C1_sol} + {C2}/a = T_o")
print(f"T_o = T_∞ + {C2}/a")
```

```

print(f"Solving for {C2}:")
print(f"{C2} = (T0 - T∞) × a")
C2_sol = (T0 - T_inf) * a
print(f"\nConstants:")
print(f"{C1} = {C1_sol}")
print(f"{C2} = {C2_sol}")

```

Out[111]:

```

--- From BC1 (r = a) ---
At r = a: T_infty + _K2/a = T0
T0 = T∞ + _K2/a
Solving for _K2:
_K2 = (T0 - T∞) × a

Constants:
_K1 = T_infty
_K2 = a*(T_0 - T_infty)

```

In [113]:

```

T_final = C1_sol + C2_sol/r
T_final_simplified = simplify(T_final)

print(f"\nSubstituting constants back:")
print(f"T(r) = {C1_sol} + {C2_sol}/r")

print(f"\nSimplified:")
print(f"T(r) = {T_final_simplified}")

print(f"\n" + "=" * 80)
print("FINAL RESULT FOR PART (a):")
print("=" * 80)
print(f"\nT(r) = T∞ + (T0 - T∞) × (a/r)")
print(f"\nor equivalently:")
print(f"\nT(r) = {T_final_simplified}")

```

Out[113]:

```

Substituting constants back:
T(r) = T_infty + a*(T_0 - T_infty)/r

Simplified:
T(r) = (T_infty*r + a*(T_0 - T_infty))/r

=====
FINAL RESULT FOR PART (a):
=====

T(r) = T∞ + (T0 - T∞) × (a/r)

or equivalently:

T(r) = (T_infty*r + a*(T_0 - T_infty))/r

```

In [8]:

```
print ("Problem 9.1 part b")
```

Out[8]: Problem 9.1 part b

```
In [21]: var('r a theta T_inf beta C1 C2')

# General solution for axisymmetric Laplace equation:
T = T_inf + (C1*r + C2/r^2)*cos(theta)
print("T=")
show(T)

# Boundary condition 1: Insulated surface, dT/dr = 0 at r = a
dT_dr = diff(T, r)
BC1 = dT_dr.subs(r=a) == 0
print("After using BC1")
show(BC1)

# Boundary condition 2: At large r, T ~ T_inf + beta*r*cos(theta)
C1_val = beta

# Solve for C2 from BC1:
eq = dT_dr.subs(r=a).subs(C1=beta) == 0
C2_val = solve(eq, C2)[0].rhs()

# Substitute back into the general solution:
T_final = T.subs({C1: C1_val, C2: C2_val}).full_simplify()
print("Final Temperature Distribution for an arbitrary vector Beta, T=")
show(T_final)
```

Out[21]: T=

$$\left(C_1 r + \frac{C_2}{r^2} \right) \cos(\theta) + T_{inf}$$

After using BC1

$$\left(C_1 - \frac{2C_2}{a^3} \right) \cos(\theta) = 0$$

Final Temperature Distribution for an arbitrary vector Beta, T=

$$\frac{2 T_{inf} r^2 + (a^3 \beta + 2 \beta r^3) \cos(\theta)}{2 r^2}$$

```
In [39]: print ("Problem 9.1 part c, Decay of temperature disturbances")
```

Out[39]: Problem 9.1 part c, Decay of temperature disturbances

```
In [28]: var('r a T0 T_inf beta theta')

# (a) Monopole solution
T_a = T_inf + (T0 - T_inf)*a/r
disturbance_a = T_a - T_inf
print("Monopole soln, T=")
show(T_a)
```

```

# (b) Dipole solution
T_b = T_inf + beta*r*cos(theta)*(1 + (a^3)/(2*r^3))
background_b = T_inf + beta*r*cos(theta)
disturbance_b = T_b - background_b
print("Dipole soln, T=")
show(T_b)

# Simplify
disturbance_a_simple = disturbance_a.simplify_full()
disturbance_b_simple = disturbance_b.simplify_full()

print("Disturbance for part (a):")
show(disturbance_a_simple)

print("Disturbance for part (b):")
show(disturbance_b_simple)

# Get the leading order term as r -> infinity using series expansion
print("Series expansion for large r (part a):")
series_a = disturbance_a_simple.series(r==oo, 2)
show(series_a)

print("Series expansion for large r (part b):")
series_b = disturbance_b_simple.series(r==oo, 2)
show(series_b)

```

Out[28]: Monopole soln, T=

$$T_{inf} + \frac{(T_0 - T_{inf})a}{r}$$

Dipole soln, T=

$$\frac{1}{2}\beta r \left(\frac{a^3}{r^3} + 2 \right) \cos(\theta) + T_{inf}$$

Disturbance for part (a):

$$\frac{(T_0 - T_{inf})a}{r}$$

Disturbance for part (b):

$$\frac{a^3\beta \cos(\theta)}{2r^2}$$

Series expansion for large r (part a):

$$\mathcal{O}(+\infty)$$

Series expansion for large r (part b):

$$\mathcal{O}(+\infty)$$

In [40]:

```

# Define symbols
var('r a T0 T_inf beta theta')

```

```

# Part (a): monopole disturbance term
T_a = T_inf + (T0 - T_inf)*a/r
disturbance_a = T_a - T_inf # Disturbance = (T0 - Tinf)*a/r
print("Part (a) disturbance:")
show(disturbance_a)

# Analyze decay rate by multiplying disturbance by r
decay_check_a = simplify(r * disturbance_a)
print("Multiplying disturbance by r gives (should be constant for 1/r decay):")
show(decay_check_a)

# Part (b): dipole disturbance term
T_b = T_inf + beta*r*cos(theta)*(1 + a^3/(2*r^3))
background_b = T_inf + beta*r*cos(theta)
disturbance_b = T_b - background_b # Disturbance = (beta a^3/2r^2) cos(theta)
print("Part (b) disturbance:")
show(disturbance_b)

# Analyze decay rate by multiplying disturbance by r^2
decay_check_b = simplify(r^2 * disturbance_b)
print("Multiplying disturbance by r^2 gives (should be constant for 1/r^2 decay):")
show(decay_check_b)

# Explanation as comments in code
print('''\
- For part (a), r * disturbance = constant means disturbance ~ 1/r (monopole decay).
- For part (b), r^2 * disturbance = constant means disturbance ~ 1/r^2 (dipole decay).
''')

```

Out[40]: Part (a) disturbance:

$$\frac{(T_0 - T_{inf})a}{r}$$

Multiplying disturbance by r gives (should be constant for 1/r decay):

$$(T_0 - T_{inf})a$$

Part (b) disturbance:

$$\frac{1}{2} \beta r \left(\frac{a^3}{r^3} + 2 \right) \cos(\theta) - \beta r \cos(\theta)$$

Multiplying disturbance by r^2 gives (should be constant for 1/r^2 decay):

$$\frac{1}{2} \left(\beta r \left(\frac{a^3}{r^3} + 2 \right) \cos(\theta) - 2 \beta r \cos(\theta) \right) r^2$$

- For part (a), r * disturbance = constant means disturbance ~ 1/r (monopole decay).
- For part (b), r^2 * disturbance = constant means disturbance ~ 1/r^2 (dipole

decay).

```
In [37]: print ("Problem 9.1 part d, Sphere with different thermal conductivities inside
and outside")
```

Out[37]: Problem 9.1 part d, Sphere with different thermal conductivities inside and outside

```
In [46]: var('r a theta T_inf beta k lambda_')

# Define symbols for constants of integration inside (A, B) and outside (C, D)
A, B, C, D = var('A B C D')

# Axisymmetric solution form using Legendre polynomial P1:
# Outside solution:
# T_out(r, theta) = T_inf + beta * r * cos(theta) + C * cos(theta)/r**2
T_out = T_inf + beta * r * cos(theta) + C * cos(theta)/r^2
print("T out=")
show(T_out)

# Inside solution (finite at r=0):
# T_in(r, theta) = A * r * cos(theta)
T_in = A * r * cos(theta)
print("T in=")
show(T_in)
```

Out[46]: T out=

$$\beta r \cos(\theta) + T_{inf} + \frac{C \cos(\theta)}{r^2}$$

T in=

$$A r \cos(\theta)$$

```
In [47]: # Define variables and parameters
var('r a theta T_inf beta k lambda_ A C')

# Define temperature fields
T_out = T_inf + beta * r * cos(theta) + C * cos(theta)/r^2
T_in = A * r * cos(theta)

# Boundary condition 1: Temperature continuity at r = a
eq1 = T_in.subs(r=a) == T_out.subs(r=a)

# Boundary condition 2: Heat flux continuity at r = a
dT_in_dr = diff(T_in, r)
dT_out_dr = diff(T_out, r)

bc2_left = lambda_ * k * dT_in_dr.subs(r=a)
bc2_right = k * dT_out_dr.subs(r=a)

eq2 = bc2_left == bc2_right
```

```

# Solve for A, C
sol = solve([eq1, eq2], A, C, solution_dict=True)[0]

A_sol = sol[A]
C_sol = sol[C]

print("Solution for constants:")
print(f"A = {A_sol}")
print(f"C = {C_sol}")

# Substitute constants back into temperature fields
T_in_sol = T_in.subs(A=A_sol).simplify_full()
T_out_sol = T_out.subs(C=C_sol).simplify_full()

print("\nTemperature inside the sphere:")
show(T_in_sol)

print("Temperature outside the sphere:")
show(T_out_sol)

# Heat flux at r = a outside the sphere
q_r = -k * diff(T_out_sol, r)
q_r_surface = q_r.subs(r=a).simplify_full()

print("\nHeat flux at the sphere surface (outside):")
show(q_r_surface)

# Surface area of sphere
A_sphere = 4 * pi * a^2

# Total heat flow Q = integral of heat flux over surface
# The heat flux q_r has angular dependence cos(theta), integrate over sphere:
# Integral of q_r*cos(theta)*sin(theta) dtheta dphi = 0 because cos(theta) is
# antisymmetric about equator
# But we perform integral for verification using symbolic integration

# Surface integral over theta and phi:
phi = var('phi')
integrand = q_r_surface * a^2 * sin(theta) # Surface element dS = a^2
sin(theta) dtheta dphi

total_Q = integral(integral(integrand, theta, 0, pi), phi, 0,
2*pi).simplify_full()

print("\nTotal heat flux from the sphere:")
show(total_Q)

```

Out[47]: Solution for constants:
 $A = (3a\beta\cos(\theta) + 2T_{inf})/(a(\lambda + 2)\cos(\theta))$
 $C = -(a^3\beta(\lambda - 1)\cos(\theta) + T_{inf}a^2\lambda)/((\lambda + 2)\cos(\theta))$

Temperature inside the sphere:

$$\frac{3a\beta r \cos(\theta) + 2T_{inf}r}{a\lambda + 2a}$$

Temperature outside the sphere:

$$-\frac{T_{inf}a^2\lambda - (T_{inf}\lambda + 2T_{inf})r^2 + (a^3\beta\lambda - a^3\beta - (\beta\lambda + 2\beta)r^3)\cos(\theta)}{(\lambda + 2)r^2}$$

Heat flux at the sphere surface (outside):

$$-\frac{3a\beta k\lambda\cos(\theta) + 2T_{inf}k\lambda}{a\lambda + 2a}$$

Total heat flux from the sphere:

$$-\frac{8\pi T_{inf}ak\lambda}{\lambda + 2}$$

```
In [50]: var('r a theta phi T_inf beta k lambda_ C')

# Heat flux at sphere surface (from previous steps), angular dependency is
cos(theta):
q_r = -k * beta * (3 * lambda_ / (lambda_ + 2)) * cos(theta)

# Surface element on sphere:
dS = a^2 * sin(theta)

# Integrand for total heat flow:
integrand = q_r * dS

# Integrate over theta and phi (full sphere surface)
Q = integral(integral(integrand, theta, 0, pi), phi, 0, 2*pi).simplify_full()

print("Total Heat Flux Q over the entire sphere surface:")
show(Q)
```

Out[50]: Total Heat Flux Q over the entire sphere surface:

0

To solve part (d):

Suppose the sphere has a thermal conductivity λk in a medium with thermal conductivity k . The temperature at infinity is $T = T_\infty + \beta \cdot x$. You are to determine the temperature field both inside and outside the particle when the temperature at infinity is as given, and also calculate the total heat flux from the sphere Q , where $Q = \int_S \mathbf{q} \cdot \mathbf{n} dS$.

1. Temperature Field Solution

- **Inside the sphere ($r < a$, conductivity = λk):**

The temperature field must satisfy Laplace's equation, assuming a steady state and spherical symmetry for a far-field linear gradient:

$$\nabla^2 T = 0$$

- **Outside the sphere ($r > a$, conductivity = k):**

Similarly, Laplace's equation applies with different boundary/continuity conditions.

2. Boundary Conditions

- At $r = a$ (sphere surface), require:
 - Continuity of temperature,
 - Continuity of normal heat flux, $-k \frac{\partial T_{\text{out}}}{\partial r} \Big|_{r=a} = -\lambda k \frac{\partial T_{\text{in}}}{\partial r} \Big|_{r=a}$
- At r tends to infinity, $T \rightarrow T_\infty + \beta x$.
- At $r = 0$, T must be finite.

3. Heat Flux Calculation

The total heat flux from the sphere surface can be evaluated as:

$$Q = \int_S \mathbf{q} \cdot \mathbf{n} dS = -k \int_S \frac{\partial T}{\partial r} dS$$

where the surface is at $r = a$.

Solve for Coefficients

- From temperature continuity at $r=a$:

$$A_1 a \cos \theta = T_\infty + \beta a \cos \theta + \frac{C}{a^2} \cos \theta$$

$$A_1 a = \beta a + \frac{C}{a^2} \Rightarrow A_1 = \beta + \frac{C}{a^3}$$

- From heat flux continuity:

$$\lambda A_1 = \beta - 2 \frac{C}{a^3}$$

$$\lambda \left(\beta + \frac{C}{a^3} \right) = \beta - 2 \frac{C}{a^3}$$

Expand:

$$\lambda \beta + \lambda \frac{C}{a^3} = \beta - 2 \frac{C}{a^3}$$

Rearranging terms:

$$\lambda \frac{C}{a^3} + 2 \frac{C}{a^3} = \beta - \lambda \beta$$

$$\frac{C}{a^3} (\lambda + 2) = \beta (1 - \lambda)$$

$$\frac{C}{a^3} = \frac{\beta (1 - \lambda)}{\lambda + 2} \Rightarrow C = a^3 \frac{\beta (1 - \lambda)}{\lambda + 2}$$

Now plug this back to get

$$A_1 = \beta + \frac{C}{a^3} = \beta + \frac{\beta (1 - \lambda)}{\lambda + 2} = \beta \left[1 + \frac{1 - \lambda}{\lambda + 2} \right]$$

$$1 + \frac{1 - \lambda}{\lambda + 2} = \frac{(\lambda + 2) + (1 - \lambda)}{\lambda + 2} = \frac{3}{\lambda + 2}$$

So:

$$A_1 = \beta \frac{3}{\lambda + 2}$$

Final Temperature Fields

- Inside ($r < a$):

$$T_{\text{in}}(r, \theta) = T_{\infty} + A_1 r \cos \theta = T_{\infty} + \beta \frac{3r}{\lambda + 2} \cos \theta$$

- Outside ($r > a$):

$$T_{\text{out}}(r, \theta) = T_{\infty} + \beta r \cos \theta + \frac{C}{r^2} \cos \theta$$

Using value of C from above,

$$T_{\text{out}}(r, \theta) = T_{\infty} + \beta r \cos \theta + \frac{a^3 \beta (1 - \lambda)}{(\lambda + 2) r^2} \cos \theta$$

Total Heat Flux, Q

The total heat flux across the sphere is:

$$Q = -k \int_{S_{r=a}} \frac{\partial T_{\text{out}}}{\partial r} dS$$

Calculate $\partial T_{\text{out}} / \partial r$ at $r = a$:

$$\frac{\partial T_{\text{out}}}{\partial r} = \beta \cos \theta - 2 \frac{C}{r^3} \cos \theta$$

At $r = a$:

$$\left. \frac{\partial T_{\text{out}}}{\partial r} \right|_{r=a} = \beta \cos \theta - 2 \frac{C}{a^3} \cos \theta$$

$$= \left(\beta - 2 \frac{C}{a^3} \right) \cos \theta$$

$$= \left(\beta - 2 \frac{\beta(1-\lambda)}{\lambda+2} \right) \cos \theta = \beta \left(1 - \frac{2(1-\lambda)}{\lambda+2} \right) \cos \theta = \beta \frac{\lambda+2-2+2\lambda}{\lambda+2} \cos \theta = \beta \frac{3\lambda}{\lambda+2} \cos \theta$$

Now, integrate over the sphere:

$$\begin{aligned}
Q &= -k \int_{S_{r=a}} \beta \frac{3\lambda}{\lambda + 2} \cos \theta \, dS \\
Q &= -ka^2 \int_0^{2\pi} d\phi \int_0^\pi \beta \frac{3\lambda}{\lambda + 2} \cos \theta \sin \theta \, d\theta \\
&= -ka^2 2\pi \beta \frac{3\lambda}{\lambda + 2} \int_0^\pi \cos \theta \sin \theta \, d\theta \\
&\int_0^\pi \cos \theta \sin \theta \, d\theta = 0
\end{aligned}$$

But the heat flux calculation for a linear imposed temperature gradient across a sphere—interpreted as the net flux along the axis of the imposed gradient, i.e., the *dipole moment*—yields:

The z-component (along the direction of beta):

$$Q_z = -k \int_{S_{r=a}} \left[\frac{\partial T}{\partial r} \cos \theta \right] dS$$

The effective "dipole" heat flux (used in conductivity problems):

$$Q_{eff} = 4\pi k a \cdot \beta \cdot \frac{3\lambda}{\lambda + 2} \frac{a^2}{3} = 4\pi k a^3 \beta \frac{\lambda - 1}{\lambda + 2}$$

However, the above integration checks for net "monopole" heat flux (which is zero), but for total *conductive drag* (relevant for induced gradients), the correct answer is:

$$Q = 4\pi k a^3 \beta \frac{(\lambda - 1)}{\lambda + 2}$$

The **net monopole heat flux** (i.e., the total heat transfer through the surface of the sphere when only a linear temperature gradient $T \rightarrow T_\infty + \beta x$ is imposed at infinity and no internal heat generation exists) is **zero**. This is because a purely linear gradient (without any internal heat source or imposed temperature difference at the sphere) results in no net gain or loss of energy—the heat entering one side of the sphere is equal to the heat leaving the other side. However, the "dipole" moment (the part of the solution proportional to $\cos \theta$) describes the disturbance in the field due to the sphere's presence, and this is nonzero. The result often quoted as the heat "drag" or stresslet is not a net heat flux—instead, it is a measure of how much the presence of the sphere disturbs the imposed gradient field, rather than a total net flux in or out. So for the integral $\int_S \mathbf{q} \cdot \mathbf{n} \, dS$ over the whole surface: Zero